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Syringe adapter

Abstract

A syringe adapter includes a female connector, a body in fluid communication with the female connector, a male connector in fluid communication with the female connector and the body, a passage extending radially through the body and a cannula extending axially through the body and out through the female connector, a first end of the cannula fluidly coupled to an end of the passage disposed within the body. A filter housing containing a filter is fluidly coupled to an end of the passage that is exterior to the body, the filter housing terminating in a vent that is fluidly coupled to the passage. A first fluid path is formed through the cannula, the passage and the vent, and a second fluid path is formed through the female connector, the body, and the male connector, wherein a closed system is provided when the male connector is uncoupled from the medical device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/674,639, entitled “SYRINGE ADAPTER,” filed on Feb. 17, 2022, which issued as U.S. Pat. No. 11,992,654 on May 28, 2024, which is a continuation of U.S. patent application Ser. No. 16/158,179, entitled “SYRINGE ADAPTER,” filed on Oct. 11, 2018, which issued as U.S. Pat. No. 11,285,261 on Mar. 29, 2022, the entire content of which is incorporated by referenced herein.

BACKGROUND

(1) The present disclosure relates generally to medical connectors used in fluid transfer applications. More particularly, the present disclosure relates to a syringe adapter for the transfer of fluids in medical settings.

(2) Medical connectors are widely used to transmit, prepare, and deliver medical fluids. The delivery of a medical fluid may include administering the medical fluid intravenously through an intravenous (IV) set from a fluid source such as a syringe.

(3) Unlike other fluid sources, such as collapsible bags, the fluid-filled syringe includes a rigid structure and cannot collapse when the plunger of the syringe is stationary and the fluid is being pulled from the syringe. As a result, ambient air needs to be introduced into the syringe to replace the fluid that is being pulled from the syringe. A traditional approach to allow air to enter the syringe, as fluid is being pulled from the syringe, includes attaching a device between the syringe and the IV set. For example, the device conventionally includes an open female Luer with a cannula that protrudes past the female Luer. The cannula is inserted into the syringe, such that a filter at the other end of the cannula provides an air path into the syringe through the cannula. When the syringe is empty and needs to be removed and replaced with another syringe, however, the female Luer will be exposed to the atmosphere, due to the cannula protruding past the female Luer, creating an open system during syringe changes.

SUMMARY

(4) One or more aspects of the present disclosure provides a syringe adapter for coupling with a syringe. The syringe adapter comprises a female connector; a body in fluid communication with the female connector; a male connector in fluid communication with the female connector and the body, the male connector configured to couple with a medical device; a passage extending radially through the body; a cannula extending axially through the body and out through the female connector, a first end of the cannula fluidly coupled to an end of the passage disposed within the body; a filter housing fluidly coupled to an end of the passage that is exterior to the body, the filter housing terminating in a vent that is fluidly coupled to the passage; and a filter disposed in the filter housing, wherein a first fluid path is formed through the cannula, the passage and the vent, and a second fluid path is formed through the female connector, the body, and the male connector, and wherein a closed system is provided when the male connector is uncoupled from the medical device.

(5) One or more aspects of the present disclosure provide a syringe adapter for coupling a syringe to a fluid system. The syringe adapter comprises a housing; a needle-free connector disposed in the housing; a communication member in slidable engagement with the housing, the communication member slidable, with respect to the housing, between an unactuated state and an actuated state of the syringe adapter; a cannula extending from the communication member, wherein, in the unactuated state, the cannula is retracted within the needle-free connector and, in the actuated state, the cannula protrudes through and axially beyond the needle-free connector; and a conduit extending from, and disposed through, the communication member, wherein the conduit is fluidly separated from the needle-free connector when the syringe adapter is coupled with the syringe and in the unactuated state.

(6) Additional features and advantages of the subject technology will be set forth in the description below, and in part will be apparent from the description, or may be learned by practice of the subject technology. The advantages of the subject technology will be realized and attained by the structure particularly pointed out in the written description and claims hereof as well as the appended drawings.

(7) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are intended to provide further explanation of the subject technology as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are included to provide further understanding of the subject technology and are incorporated in and constitute a part of this description, illustrate aspects of the subject technology and, together with the specification, serve to explain principles of the subject technology.

(2) FIG. 1 illustrates a schematic view of fluid delivery system in accordance with aspects of the present disclosure.

(3) FIG. 2 illustrates a side view of a syringe adapter depicted in an uncoupled, unactuated state, with portions sectioned and broken away to show details, in accordance with aspects of the present disclosure.

(4) FIG. 3 illustrates a side view of the syringe adapter of FIG. 2 depicted in the uncoupled, unactuated state with a syringe in close proximity, but uncoupled to the syringe adapter, in accordance with aspects of the present disclosure.

(5) FIG. 4 illustrates a side view of the syringe adapter of FIG. 2 depicted in a coupled, unactuated state with the syringe coupled to the syringe adapter, in accordance with aspects of the present

disclosure.

(6) FIG. 5 illustrates a side view of the syringe adapter of FIG. 2 depicted in a coupled, actuated state with the syringe coupled to the syringe adapter, in accordance with aspects of the present disclosure.

(7) FIG. 6 illustrates a detail side view of the syringe adapter of FIG. 4, in accordance with aspects of the present disclosure.

(8) FIG. 7 illustrates a detail side view of the syringe adapter of FIG. 5, in accordance with aspects of the present disclosure.

(9) FIG. 8 illustrates a perspective view of an alternative embodiment of a syringe adapter, in accordance with aspects of the present disclosure.

(10) FIG. 9 illustrates a cross-sectional perspective view of the syringe adapter of FIG. 8, in accordance with aspects of the present disclosure.

(11) FIG. 10 illustrates a side view of the syringe adapter of FIG. 8 attached to a syringe, in accordance with aspects of the present disclosure.

(12) FIG. 11 illustrates a cross-sectional side view of the syringe adapter attached to the syringe in FIG. 10, in accordance with aspects of the present disclosure.

(13) FIG. 12 illustrates a perspective view of another alternative embodiment of a syringe adapter, in accordance with aspects of the present disclosure.

(14) FIG. 13 illustrates a perspective view of the syringe adapter of FIG. 12 attached to a syringe, in accordance with aspects of the present disclosure.

(15) FIG. 14 illustrates a cross-sectional view of an air vent of the syringe adapter in FIG. 12, in accordance with aspects of the present disclosure.

(16) FIG. 15 illustrates a detail side view of the syringe adapter of FIG. 12 attached to the syringe, with portions sectioned and broken away to show internal fluid paths, in accordance with aspects of the present disclosure.

(17) FIG. 16 illustrates a perspective view of another alternative embodiment of a syringe adapter depicting a container in a pre-filled state, in accordance with aspects of the present disclosure.

(18) FIG. 17 illustrates a perspective view of the syringe adapter of FIG. 16 depicting the container in a filled state, in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

(19) In the following detailed description, specific details are set forth to provide an understanding of the subject technology. It will be apparent, however, to one ordinarily skilled in the art that the subject technology may be practiced without some of these specific details. In other instances, well-known structures and techniques have not been shown in detail so as not to obscure the subject technology.

(20) A phrase such as “an aspect” does not imply that such aspect is essential to the subject technology or that such aspect applies to all configurations of the subject technology. A disclosure relating to an aspect may apply to all configurations, or one or more configurations. An aspect may provide one or more examples of the disclosure. A phrase such as “an aspect” may refer to one or more aspects and vice versa. A phrase such as “an embodiment” does not imply that such embodiment is essential to the subject technology or that such embodiment applies to all configurations of the subject technology. A disclosure relating to an embodiment may apply to all embodiments, or one or more embodiments. An embodiment may provide one or more examples of the disclosure. A phrase such as “an embodiment” may refer to one or more embodiments and vice versa. A phrase such as “a configuration” does not imply that such configuration is essential to the subject technology or that such configuration applies to all configurations of the subject technology. A disclosure relating to a configuration may apply to all configurations, or one or more configurations. A configuration may provide one or more examples of the disclosure. A phrase such as “a configuration” may refer to one or more configurations and vice versa.

(21) The various embodiments of a syringe adapter illustrated in FIGS. 1-17 are configured to

couple with a fluid source and provide a closed system, including a closed sealed surface, during a fluid source change from an IV set. In certain embodiments, in addition to including a closed sealed surface during fluid source changes from the IV set, the syringe adapter also includes a cannula in fluid communication with an air vent to allow filtered air to enter the fluid source as fluid is pulled out of the fluid source. In such embodiments, the syringe adapter is configured to prevent fluid in the fluid source from contacting a filter disposed in the air vent.

(22) FIG. 1 illustrates an exemplary fluid delivery system **100**. The fluid delivery system **100** includes a syringe **110**, a syringe adapter **112**, and an intravenous (IV) set **114**. The fluid delivery system **100** is an exemplary system in which the syringe adapter **112** is utilized and it is understood that the syringe adapter **112** may be used in other fluid delivery systems. The syringe **110** includes a syringe barrel **116**, a collar **118** extending radially outwardly from the syringe barrel **116** at one end of the syringe barrel **116**, and an end wall **120** formed at an opposite end of the syringe barrel **116**. The syringe **110** also includes a plunger **122**. The plunger **122** includes a body **124**, a thumb press **126** attached to one end of the body **124**, and a stopper **128** (shown in phantom) attached to an opposite end of the body **124**. The syringe barrel **116** slidably receives the stopper **128** and the body **124** of the plunger **122**, such that a syringe chamber **130** is formed within the syringe barrel **116** between the stopper **128** and the end wall **120** of the syringe barrel **116**. A tip **132** extends axially outwardly from the end wall **120** and is in fluid communication with the syringe chamber **130**. In some aspects, the tip **132** is a male Luer connector.

(23) The syringe adapter **112** is configured to matingly couple with the tip **132**. The syringe adapter **112** is also configured to matingly couple with a connector **134** of the IV set **114**. The IV set **114** includes tubing **136** connected to a fitting **138**. For example, the fitting **138** is a needle-free Luer connector suitable for connection to an infusion device (not shown), such as an IV needle. In some aspects, a pumping segment **140**, which is a section of tubing suitable for peristaltic manipulation to cause fluid to flow through the tubing and includes alignment fittings **142** to facilitate proper placement of the pumping segment **140** in a pump (not shown), connects segments of the tubing **136**. The IV set **114** also includes a clamp **144** that can be closed to stop flow through the tubing **136**.

(24) FIGS. 2-7 illustrate an embodiment of a syringe adapter **200**. In some aspects, the syringe adapter **200** includes a housing **210** slidably engaged with a communication member **212**. The housing **210** includes a connection port **214** disposed at one end of the housing **210** and a receiving opening **216** disposed at the other end of the housing **210**. A female connector **218** forms part of the housing **210** adjacent the connection port **214**. The female connector is configured to couple with male connectors on various medical device components. In some aspects, the female connector is a female Luer connector configured to couple with male Luer connectors on various medical device components. The housing **210** also includes a needle-free connector section **220** and a receiving section **222**, such that the needle-free connector section **220** is disposed between the connection port **214** and the receiving section **222**. The receiving section **222** is disposed between the needle-free connector section **220** and the receiving opening **216**.

(25) A needle-free connector **224** is disposed at the needle-free connector section **220** of the housing **210**. The needle-free connector **224** includes the connection port **214** and the female connector **218**. The needle-free connector **224** also includes a hollow cylinder **226** disposed within the needle-free connector section **220** of the housing **210**. The hollow cylinder **226** includes a head **228** at one end and a base **230** at the other end. The head **228** of the hollow cylinder **226** is offset axially inwardly from the connection port **214** within the needle-free connector section **220** of the housing **210**, such that a bore **232** extends from the connection port **214** to the head **228**. The connection port **214**, the bore **232**, and the hollow cylinder **226** are all coaxially aligned.

(26) The needle-free connector **224** also includes a resiliently deformable piston element **234**. In an uncoupled, unactuated state of the syringe adapter **200** shown in FIG. 2, the resiliently deformable piston element **234** is captured between the connection port **214** and the base **230**. The piston

element **234** includes a piston **236** and a compressible member **238** formed to the piston **236**. In some aspects, the compressible member **238** includes a plurality of bellows **240**. The piston **236** includes a piston head **242**. The piston head **242** includes a scalable orifice **244** disposed therethrough, which is in a sealed state when the piston element **234** is uncompressed and is in an opened state when the piston element **234** is compressed. In the uncoupled, unactuated state of the syringe adapter **200**, as depicted in FIGS. **2** and **3**, the piston element **234** is uncompressed with the scalable orifice **244** in a sealed state, such that the compressible member **238** is urged against the base **230** while the piston head **242** is located internally to the housing **210** within the bore **232** and is sealingly flush with the surrounding connection port **214** collectively forming a flat, closed, and swabable surface **245**. Moreover, the piston element **234** is hollow and includes a piston passage **246**, which extends from an exit orifice **247** of the compressible member **238** to the scalable orifice **244**.

(27) The receiving section **222** includes a transition section **248** disposed adjacent to the needle-free connector section **220**. The transition section **248** includes an annular shoulder **250** axially offset from the base **230**, such that a chamber **252** is formed between the base **230** and the shoulder **250**. The chamber **252** is fluidly coupled to the piston passage **246** via the exit orifice **247** and is coaxially aligned with an aperture **253** disposed in the shoulder **250**. A channel **254** is disposed in the receiving section **222** and includes a closed end **256**, which is disposed adjacent the chamber **252** in the transition section **248**. The channel **254** includes a transfer passage **258** disposed proximate the closed end **256**. The channel **254** is fluidly coupled to the chamber **252** via the transfer passage **258**. The channel **254** extends axially from the closed end **256** through the shoulder **250** to an opened end **260**. The opened end **260** of the channel **254** terminates within the receiving section **222** and is axially offset from the receiving opening **216** of the housing **210**. The receiving section **222** also includes a slot **262** that extends axially between the shoulder **250** and the receiving opening **216**. The slot **262** is configured to slidably guide a neck **264** of the communication member **212** when the syringe adapter **200** transitions from a coupled, unactuated state to a coupled, actuated state.

(28) With particular reference to FIGS. **2** and **6-7**, the communication member **212** includes the neck **264**, a cannula **266**, and a conduit **268**. The neck **264** extends radially from the communication member **212** and terminates at a valve housing **270** that includes an inlet port **272**. A duct **274** is disposed within the communication member **212** through the neck **264** and the valve housing **270** to the inlet port **272**. A filter housing **276** is received in fluid communication by the inlet port **272**. The filter housing **276** surrounds a filter **278** and includes at least one vent **280** configured to allow ambient air to pass through the filter **278**. A valve **282** is disposed in the valve housing **270** and is configured to allow fluid to flow from the at least one vent **280** through the filter **278** and into the duct **274** and prevent fluid to flow in the opposite direction from the duct **274** to the filter housing **276** ensuring that the filter **278** will not get wetted from fluid possibly in the duct **274**.

(29) The cannula **266** extends axially from, and through, the communication member **212** and is fluidly coupled to the duct **274** at one end and terminates at an opening **267** at an opposite end. The conduit **268** is radially offset from the cannula **266** and extends axially from, and through, the communication member **212**. Externally of the communication member **212**, a crown **284** is disposed at the end of the conduit **268**, such that the conduit **268** terminates axially at the crown **284** to fluidly couple with a passage **286**. A first O-ring **288** is disposed around the crown **284** and a second O-ring **290** is disposed around the conduit **268** between the passage **286** and the communication member **212**. Internally of the communication member **212**, the conduit **268** is fluidly coupled to an exit port **292**. A connector **294** is disposed around the exit port **292** and is configured to couple with various medical devices, such as, for example, the connector **134** of the IV set **114**.

(30) Upon assembly of the syringe adapter **200**, the communication member **212** is received, in

slidably engagement, by the receiving opening **216** of the housing **210**, such that the conduit **268** is slidably received by the channel **254**, the cannula **266** is slidably received by a third O-ring **296** disposed in the aperture **253**, and the neck **264** is slidably received by the slot **262**. The third O-ring **296** is configured to seal the cannula **266** with respect to the aperture **253** while allowing the cannula **266** to sealably slide along within the O-ring **296**.

(31) FIG. 3 illustrates the syringe adapter **200** coupled to an IV set, such as, for example, the IV set **114** and in an uncoupled, unactuated state. A pre-filled syringe, such as, for example, the syringe **110** is in close proximity to, and prepared for coupling with, the syringe adapter **200**. In the uncoupled, unactuated state, the cannula **266** is retracted into the needle-free connector **224**. For example, the piston element **234** is uncompressed, such that the opening **267** of the cannula **266** is arranged in the piston passage **246** within the compressible member **238**, the neck **264** rests in the slot **262** proximate the receiving opening **216**, and the conduit **268** is arranged in the channel **254** proximate the opened end **260**, such that the passage **286** is unaligned with the transfer passage **258** and the conduit **268** is fluidly separated from the needle-free connector **224**. Moreover, the scalable orifice **244** is in the sealed state with the piston head **242** sealingly flush with the surrounding connection port **214** forming the flat, closed, and swabable surface **245**.

(32) FIG. 4 illustrates the syringe adapter **200** in a coupled, unactuated state with the syringe **110**. With the syringe **110** coupled to the syringe adapter **200**, the male Luer tip **132** is in contact with the piston head **242** and compresses the piston element **234** to move the scalable orifice **244** from the sealed state to the opened state. In the coupled, unactuated state, the neck **264** and the conduit **268** are arranged similarly as in the uncoupled, unactuated state, but with the piston element **234** in the compressed state, the opening **267** of the cannula **266** is now arranged in the piston passage **246** within the piston **236** instead of the compressible member **238**. As the passage **286** and the transfer passage **258** are unaligned in this state and with the first O-ring **288** in sealing contact with the channel **254**, ambient air from the filter housing **276** is prevented from flowing through the channel **254** and past the first O-ring **288** to the IV set **114** via the exit port **292**. This is especially beneficial in scenarios where the IV set **114** includes the pumping segment **140** inserted into a pump and the pump is turned on before the syringe adapter **200** transitions to the coupled, actuated state. Moreover, if the plunger **122** is accidentally depressed in this state, the valve **282** prevents fluid from the syringe chamber **130** from flowing to the filter **278**.

(33) In order to allow fluid to flow from the syringe **110** to the IV set **114**, the syringe adapter **200** is required to enter the coupled, actuated state, as illustrated in FIG. 5. For the syringe adapter **200** to enter the coupled, actuated state from the coupled, unactuated state, the housing **210** is slid axially around, and locked to, the communication member **212**, such that the cannula **266** is protruded from the needle-free connector **224**. For example, the neck **264** is arranged in the slot **262** proximate the shoulder **250**, the communication member **212** abuts the shoulder **250**, the conduit **268** is arranged in the channel **254** with the crown **284** abutting the closed end **256** of the channel **254**, and the opening **267** of the cannula **266** is arranged in the syringe **110** axially beyond the end wall **120** at a location that is axially beyond a fluid intake at the needle-free connector **224**. With the crown **284** abutting the closed end **256** of the channel **254**, the passage **286** aligns with the transfer passage **258** to fluidly couple the conduit **268** with the chamber **252** and to the needle-free connector **224**. The second O-ring **290** seals the conduit **268** with the channel **254** and prevents fluid entering the passage **286** from traveling axially past the second O-ring **290** into the channel **254**.

(34) When the syringe adapter **200** is in the coupled, actuated state a first fluid path is formed, at least, through the at least one vent **280**, the duct **274**, and the cannula **266**. For example, ambient air enters the first fluid path through the at least one vent **280**, the filter **278**, the valve **282**, the duct **274**, the cannula **266**, and the opening **267** of the cannula **266** into the syringe chamber **130**. As the first fluid path provides filtered ambient air into the syringe chamber **130**, a reservoir is created in the syringe chamber **130** allowing the fluid in the syringe chamber **130** to vent or flow through a

second fluid path without requiring the plunger **122** to be depressed. The second fluid path is formed, at least, through the connection port **214**, the conduit **268**, and the exit port **294**. For example, fluid from the syringe chamber **130** flows through the tip **132** and the around the cannula **266** following the second fluid path at the connection port **214** through the bore **232**, the sealed orifice **244** in the opened state, the piston passage **246**, the exit orifice **247**, the chamber **252**, the transfer passage **258**, the passage **286**, the conduit **268**, and the exit port **294** to the IV set **114**. In some aspects, the fluid in the syringe chamber **130** flows through the second fluid path by gravity. In other aspects in which the IV set **114** includes the pumping segment **140**, a bag pump (not shown) is utilized, such that the pumping segment **140** is inserted into the bag pump to regulate peristaltic manipulation causing the fluid to flow through the second fluid path. For example, the syringe adapter **200** can be utilized with the bag pump, such as a large volume pump (LVP), in such a manner that a depleted IV bag can be replaced with the syringe **110** and syringe adapter **200** without necessitating a separate syringe pump.

(35) When the syringe **110** is ready to be replaced the syringe adapter **200** is transitioned from the coupled, actuated state to the coupled, unactuated state by sliding the housing **210** axially with respect to the communication member **212**, and then transitioned from the coupled, unactuated state to the uncoupled, unactuated state by uncoupling the syringe **110** from the syringe adapter **200**. In the uncoupled, unactuated state, the scalable orifice **244** is in the scaled state and the piston head **242** is located internally to the housing **210** within the bore **232** and is sealingly flush with the surround connection port **214**, such that closed, flat surface **245** can be swabbed before coupling a replacement syringe to the syringe adapter **200**.

(36) FIGS. **8-11** illustrate another embodiment of a syringe adapter **800**. Similar to the syringe adapter **200**, the syringe adapter **800** is couplable to a fluid source and includes a first fluid path that allows filtered air to enter the fluid source as fluid is pulled out of the fluid source through a second fluid path while preventing a filter in the first fluid path from exposure to fluid in the fluid source. The syringe adapter **800** also provides a closed system during a fluid source change from an IV set.

(37) The syringe adapter **800** includes a female connector **810**, a male connector **812**, and a body **814** fluidly coupling the female connector **810** to the male connector **812**. In some aspects, the female connector **810** is a female Luer connector and the male connector **812** is a male Luer connector. The syringe adapter **800** also includes a passage **816** extending radially through the body **814**. A filter housing **818** is fluidly coupled to an end of the passage **816** that is exterior to the body **814**. The filter housing **818** terminates at a vent **820**, which is also in fluid communication with the passage **816**. A filter **822** is disposed in the filter housing **818** and filters ambient air entering the vent **820**. In some aspects, a valve (not shown) is disposed in the filter housing **818** between the filter **822** and the passage **816** and is configured to allow ambient air to flow from the vent **820** to the passage **816** via the filter **822** and to prevent fluid from flowing from the passage **816** to the filter **822**. In some aspects, the filter housing **818** is elevated with respect to the female connector **810** and is angled with respect to the passage **816** and extends axially away from the male connector **812** and axially beyond the female connector **810**, such that the filter housing **818** and the filter **822** is arranged above (e.g., axially beyond) the female connector **810**.

(38) The syringe adapter **800** also includes a cannula **824** fluidly coupled to the passage **816** at an end of the passage **816** that is disposed within the body **814**. The cannula **824** extends axially within the body **814** and out through the female connector **810**. An opening **826** of the cannula **824** is arranged axially offset from, and exterior to, the female connector **810**. A first fluid path is formed through, at least, the cannula **824**, the passage **816**, and the vent **820**. A second fluid path is formed through, at least, the female connector **810**, the body **814**, and the male connector **812**.

(39) The female connector **810** is configured to couple with a male connector of a fluid source, such as, for example, the syringe **110**. The male connector **812** is configured to couple with a female connector of a medical device, such as, for example, the IV set **114**. In operation, the

syringe adapter **800** is coupled to the IV set **114** and the syringe **110** is coupled to the syringe adapter **800**, such that the opening **826** of the cannula **824** is arranged within the syringe chamber **130** of the syringe **110**. As the first fluid path provides filter ambient air into the syringe chamber **130**, a reservoir is created in the syringe chamber **130** allowing the fluid in the syringe chamber **130** to vent through the second fluid path without requiring the plunger **122** to be depressed. Moreover, in aspects where the filter housing **818** is angled with respect to the passage **816** and extends axially away from the male connector **812** and axially beyond the female connector **810**, the filter **822** is prevented from coming in contact with the fluid from the syringe chamber **130**, if the plunger **122** is accidentally depressed, due to the arrangement of filter housing **818** with respect to fluid intake at the second fluid path. In some aspects, the fluid in the syringe chamber **130** flows through the second fluid path by gravity. In other aspects in which the IV set **114** includes the pumping segment **140**, a bag pump (not shown) is utilized, such that the pumping segment **140** is inserted into the bag pump to regulate peristaltic manipulation causing the fluid to flow through the second fluid path.

(40) When the syringe **110** is ready to be replaced the syringe adapter **800**, while still coupled to the syringe **110**, is uncoupled from the IV set **114** providing a closed system during the syringe change. Both the syringe **110** and the syringe adapter **800** can be disposed and a new syringe adapter **800** can be coupled to the IV set **114** so that a replacement syringe can be coupled to the syringe adapter **800** without having to break the closed system.

(41) FIGS. **12-15** illustrate another embodiment of a syringe adapter **1200**. Similar to the syringe adapter **200**, the syringe adapter **1200** is couplable to a fluid source and also includes a first fluid path that allows filtered air to enter the fluid source as fluid is pulled out of the fluid source through a second fluid path while preventing a filter in the first fluid path from exposure to fluid in the fluid source. The syringe adapter **1200** also provides a closed system during a fluid source change from an IV set.

(42) Similar to the syringe adapter **800**, the syringe adapter **1200** includes a female connector **1210**, a male connector **1212**, and a body **1214** fluidly coupling the female connector **1210** to the male connector **1212**. In some aspects, the female connector **1210** is a female Luer connector and the male connector **1212** is a male Luer connector. The syringe adapter **1200** includes a passage **1216** extending radially through the body **1214**. A flexible tubing **1218** is fluidly coupled to an end of the passage **1216** that is exterior to the body **1214**. The flexible tubing **1218** is also fluidly coupled to a filter housing **1220**. The filter housing **1220** terminates at a vent **1222**, which is also in fluid communication with the flexible tubing **1218**. A filter **1224** is disposed in the filter housing **1220** between the vent **1222** and the flexible tubing **1218** and filters ambient air entering the vent **1222** to the flexible tubing **1218**. In some aspects, a cap **1225** is hingedly coupled to the filter housing **1220** to secure the filter **1224** within the filter housing **1220** and facilitate in replacement of the filter **1224**.

(43) The syringe adapter **1200** also include a cannula **1226** fluidly coupled to the passage **1216** at an end of the passage **1216** that is disposed within the body **1214**. The cannula **1226** extends axially within the body **1214** and out through the female connector **1210**. An opening **1228** of the cannula **1226** is arranged axially offset from, and exterior to, the female connector **1210**. A first fluid path is formed through, at least, the cannula **1226**, the passage **1216**, the flexible tubing **1218**, and the vent **1222**. A second fluid path is formed through, at least, the female connector **1210**, the body **1214**, and the male connector **1212**.

(44) The syringe adapter **1200** is configured to couple with a fluid source, such as, for example, the syringe **110** and includes a clip **1230** for securing the flexible tubing **1218** in place against the syringe barrel **116** of the syringe **110** when the syringe adapter **1200** is coupled to the syringe **110**. In some aspects, the clip **1230** includes an annular shape with a slit **1232** to facilitate the placement of the clip **1230** around the syringe barrel **116** and the flexible tubing **1218** for holding the flexible tubing **1218** to the syringe barrel **116**. For example, when the syringe adapter **1200** is coupled to the

syringe **110**, the flexible tubing **1218** is arranged axially along the syringe barrel **116**, such that the clip **1230** is arranged around the syringe barrel **116** with the filter housing **1220** proximate the collar **118** and with the flexible tubing **1218** secured against the syringe barrel **116** and arranged between the clip **1230** and the syringe barrel **116**. With the filter housing **1220** arranged proximate the collar **118** and above (e.g., axially beyond) a level of any fluid in the syringe chamber **130**, the filter housing is elevated with respect to the female connector **1210** and the filter **1224** is prevented from contact with fluid from the syringe chamber **130** if the plunger **122** is accidentally depressed.

(45) The female connector **1210** is configured to couple with a male connector of a fluid source, such as, for example, the syringe **110**. The male connector **1212** is configured to couple with a female connector of a medical device, such as, for example, a needle-free connector **1234**, which is, in turn, coupled to the IV set **114**, for example. In operation, the syringe adapter **1200** is coupled to the IV set **114** via the needle-free connector **1234** and the syringe **110** is coupled to the syringe adapter **1200**, such that the opening **1228** of the cannula **1226** is arranged within the syringe chamber **130** of the syringe **110**. As the first fluid path provides filter ambient air into the syringe chamber **130**, a reservoir is created in the syringe chamber **130** allowing the fluid in the syringe chamber **130** to vent through the second fluid path without requiring the plunger **122** to be depressed. Further, the filter **1224** is prevented from contact with fluid from the syringe chamber **130** in scenarios when the plunger **122** is accidentally depressed because the filter housing **1220** and the filter **1224** is raised above the level of any fluid in the syringe chamber **130**. In some aspects, the fluid in the syringe chamber **130** flows through the second fluid path by gravity. In other aspects in which the IV set **114** includes the pumping segment **140**, a bag pump (not shown) is utilized, such that the pumping segment **140** is inserted into the bag pump to regulate peristaltic manipulation causing the fluid to flow through the second fluid path.

(46) FIGS. **16** and **17** illustrate another embodiment syringe adapter **1600**. The syringe adapter **1600** also provides a closed system during a fluid source change from an IV set. The syringe adapter **1600** includes a female connector **1610**, a male connector **1612**, and a body **1614** fluidly coupling the female connector **1610** to the male connector **1612**. In some aspects, the female connector **1610** is a female Luer connector and the male connector **1612** is a male Luer connector. The syringe adapter **1600** includes a chamber **1616** fluidly coupled to, and extending outwardly from, the body **1614**. The chamber **1616** can be, for example, a resilient bag, a non-resilient bag, or a collapsible plastic container. In some aspects, the syringe adapter **1600** includes a holder **1618** for supporting the chamber **1616**. The syringe adapter **1600** also includes a needle-free connector **1620** disposed at, and in fluid communication with, the female connector **1610**. In some aspects, the needle-free connector **1620** integrally formed with the female connector **1610**. In some other aspects, the needle-free connector **1620** includes a male connector, which is couplable to the female connector **1610**. The needle-free connector **1620** also includes a female fitting **1622**, which is couplable to a male connector of a medical device, such as the syringe **110**. A valve **1624** (shown in phantom), such as a check valve, for example, is disposed between the needle-free connector **1620** and the female connector **1610**. The valve **1624** is configured to allow fluid flow from the needle-free connector **1620** to the female connector **1610** and prevent fluid flow from the female connector **1610** to the needle-free connector **1620**.

(47) In operation, the male connector **1612** of the syringe adapter **1600** is coupled to a medical device, such as, for example, a pump fitting or the connector **134** of the IV set **114**. With the syringe adapter **1600** coupled to the medical device, the syringe **110** is then coupled to the needle-free connector **1620** of the syringe adapter **1600**. The plunger **122** of the syringe **110** is depressed to push the fluid in the syringe **110** through the needle-free connector **1620**, the female connector **1610**, and into the chamber **1616**. For example, in aspects where the chamber **1616** is a resilient elastomeric bag, the chamber **1616** is unfilled with fluid, as illustrated in FIG. **16**, and is inflated and filled with fluid, as illustrated in FIG. **17**. With the fluid emptied from the syringe **110** and filled in the chamber **1616**, the syringe **110** can be uncoupled from the syringe adapter **1600** to

provide a closed system via the needle-free connector **1620**. Fluid in the chamber **1616** can then subsequently flow through the male connector **1612** to the IV set **114** either by gravity or via a pump (not shown) as atmospheric pressure allows the fluid in the chamber **1616** to flow without resistance. When additional medication is required the needle-free connector **1620** can be swabbed and a new syringe coupled to the needle-free connector **1620** to fill the chamber **1616** as described above.

(48) The foregoing description is provided to enable a person skilled in the art to practice the various configurations described herein. While the subject technology has been particularly described with reference to the various figures and configurations, it should be understood that these are for illustration purposes only and should not be taken as limiting the scope of the subject technology.

(49) There may be many other ways to implement the subject technology. Various functions and elements described herein may be partitioned differently from those shown without departing from the scope of the subject technology. Various modifications to these configurations will be readily apparent to those skilled in the art, and generic principles defined herein may be applied to other configurations. Thus, many changes and modifications may be made to the subject technology, by one having ordinary skill in the art, without departing from the scope of the subject technology.

(50) As used herein, the phrase “at least one of” preceding a series of items, with the term “and” or “or” to separate any of the items, modifies the list as a whole, rather than each member of the list (i.e., each item). The phrase “at least one of” does not require selection of at least one of each item listed; rather, the phrase allows a meaning that includes at least one of any one of the items, and/or at least one of any combination of the items, and/or at least one of each of the items. By way of example, the phrases “at least one of A, B, and C” or “at least one of A, B, or C” each refer to only A, only B, or only C; any combination of A, B, and C; and/or at least one of each of A, B, and C.

(51) Furthermore, to the extent that the term “include,” “have,” or the like is used in the description or the claims, such term is intended to be inclusive in a manner similar to the term “comprise” as “comprise” is interpreted when employed as a transitional word in a claim. The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments.

(52) A reference to an element in the singular is not intended to mean “one and only one” unless specifically stated, but rather “one or more.” The term “some” refers to one or more. All structural and functional equivalents to the elements of the various configurations described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and intended to be encompassed by the subject technology. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the above description.

(53) While certain aspects and embodiments of the subject technology have been described, these have been presented by way of example only, and are not intended to limit the scope of the subject technology. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms without departing from the spirit thereof. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the subject technology.

Claims

1. A syringe adapter for coupling a syringe to a fluid system, the syringe adapter comprising: a female connector; a body in fluid communication with the female connector; a male connector in fluid communication with the female connector and the body, the male connector configured to couple with a medical device; a passage extending radially through the body; a cannula extending

axially through the body and out through the female connector, a first end of the cannula fluidly coupled to an end of the passage disposed within the body; a filter housing fluidly coupled to an end of the passage that is exterior to the body, the filter housing terminating in a vent that is fluidly coupled to the passage; and a filter disposed in the filter housing, wherein a first fluid path is formed through the cannula, the passage and the vent, and a second fluid path is formed through the female connector, the body, and the male connector, and wherein a closed system is provided when the male connector is uncoupled from the medical device.

2. The syringe adapter of claim 1, further comprising a valve disposed in the filter housing between the filter and the passage.
3. The syringe adapter of claim 2, wherein the valve is configured to allow ambient air to flow from the vent to the passage via the filter.
4. The syringe adapter of claim 2, wherein the valve is configured to prevent fluid from flowing from the passage to the filter.
5. The syringe adapter of claim 1, wherein the filter housing is elevated with respect to the female connector.
6. The syringe adapter of claim 1, wherein the filter housing is angled with respect to the passage and extends axially away from the male connector.
7. The syringe adapter of claim 1, wherein the filter housing extends axially beyond the female connector.
8. The syringe adapter of claim 1, wherein the female connector is coupled to the syringe such that a second end of the cannula is arranged within a chamber of the syringe.
9. The syringe adapter of claim 1, wherein the medical device is an intravenous set.
10. The syringe adapter of claim 1, wherein the medical device is a needle-free connector.
11. A syringe adapter for coupling with a syringe, the syringe adapter comprising: a housing; a needle-free connector disposed in the housing; a communication member in slidable engagement with the housing, the communication member slidable, with respect to the housing, between an unactuated state and an actuated state of the syringe adapter; a cannula extending from the communication member, wherein, in the unactuated state, the cannula is retracted within the needle-free connector and, in the actuated state, the cannula protrudes through and axially beyond the needle-free connector; and a conduit extending from, and disposed through, the communication member, wherein the conduit is fluidly separated from the needle-free connector when the syringe adapter is coupled with the syringe and in the unactuated state.
12. The syringe adapter of claim 11, wherein the conduit is in fluid communication with the needle-free connector when the syringe adapter is coupled with the syringe and in the actuated state.
13. The syringe adapter of claim 12, further comprising a duct extending from, and through, the communication member, wherein the duct fluidly couples the cannula to at least one vent disposed in a filter housing.
14. The syringe adapter of claim 13, further comprising a filter disposed in the filter housing and a valve disposed between the filter and the cannula, wherein the valve is configured to allow fluid to flow from the at least one vent through the filter to the cannula and is configured to prevent fluid flow from the cannula to the filter.
15. The syringe adapter of claim 14, further comprising a first fluid path and a second fluid path when the syringe adapter is coupled to the syringe and in the actuated state.
16. The syringe adapter of claim 15, wherein the first fluid path is configured to provide filtered ambient air into the syringe and the second fluid path is configured to allow fluid from the syringe to flow to an intravenous set coupled to the syringe adapter.
17. The syringe adapter of claim 16, wherein the first fluid path flows through at least the cannula, the duct, and the at least one vent.
18. The syringe adapter of claim 16, further comprising a piston element disposed in the needle-free connector, the piston element comprising a piston head and a sealable orifice disposed through

the piston head, wherein the sealable orifice is in a sealed state when the piston element is in an uncompressed state and is in an opened state when the piston element is in a compressed state.

19. The syringe adapter of claim 18, wherein the needle-free connector comprises a connection port sealingly flush with the piston head and the sealable orifice when the syringe adapter is uncoupled with the syringe.

20. The syringe adapter of claim 19, wherein the second fluid path flows through at least the connection port, the piston element, and the conduit.
